

Is the AI use of water controversy overblown?

Posted by u/queenofsevens • 0 points • 3 comments

I'm skeptical because we've had server warehouses way before AI became widely available, surely this is not a new issue--so why are we just hearing about this now?

Why would these companies need to use fresh water to cool servers? Why don't they use waste water or recycled water?

Even if they use fresh water, what happens after it leaves the warehouse? The water should enter cold and leave a little warmer. It's not contaminated at that point, cant it just move back into the freshwater system?

I understand little about servers and AI and less about plumbing. Really just trying to understand if all these claims of AI wasting thousands of gallons of water are well-founded or not.

Comments

u/Mission_Engineering8 • 3 points • 2025-05-27 04:55 UTC

Large data centers are cooled by chiller plants. The typical setup is a closed loop of water leaving the chiller, going to the data center where it absorbs heat, then it goes back to the chiller. But the chiller needs a place to get rid of that heat. So there's a second water loop that goes from the chiller to a cooling tower. The warm water is sprayed across media and a fan pulls air across it. Some of the water evaporates taking the heat with it cooling the water. The remaining water goes back to the chiller but more water is needed to make up the evaporated water.

So there's two water loops connected by a chiller. One is closed circuit while the other is open circuit where water evaporates. It's the makeup water that is a big hit. A data center can evaporate hundreds of gallons an hour every hour of every day.

The water needs to be clean "city water" or biological growth creates problems for the people exposed to the water.

u/SparkleSweetiePony • 2 points • 2025-05-27 02:34 UTC

It's overblown.

Computer cooling is generally done with liquid coolers which are closed systems, so the use is minimal in that case.

Sometimes it's done via AC systems, as pointed out by other commenter, chilled with municipal water supply, water is then returned into the water bodies via treatment.

Power plants may use lots of water, but those are also in closed or semi-closed loop systems. For example, a nuclear power plant would use a closed water loop for cooling the reactor core which then heats the other loop into steam to generate power. The runoff steam is then condensed in a heat exchanger which uses seawater, river water or cooling towers. In the case of cooling towers a small amount of water may be lost and needs refilling, but in other cases it's a closed loop.

u/PickleJuiceMartini • 1 points • 2025-05-27 00:16 UTC

A large business uses lots of water for cooling. Instead of hundreds of air conditioning (AC) units there is a large AC unit that is cooled by water. They are called chillers. This water uses clean city water. Waste water would require filtering and may have a bad smell. Some of the water evaporates during the cooling process. The remaining water is not wasted. It proceeds to the water treatment facility.